

Traffic Sign Classification

Deep Feature Manifolds & Hybrid SVM Optimization

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Introduction

Project Objectives

- **Goal:** Build a robust traffic sign recognition system for autonomous vehicles.
- **Key Approach:** Bridge the gap between Deep Learning and Support Vector Machines.
- **Innovation:** Investigation of Low-Dimensional (3D) Manifold Classification.

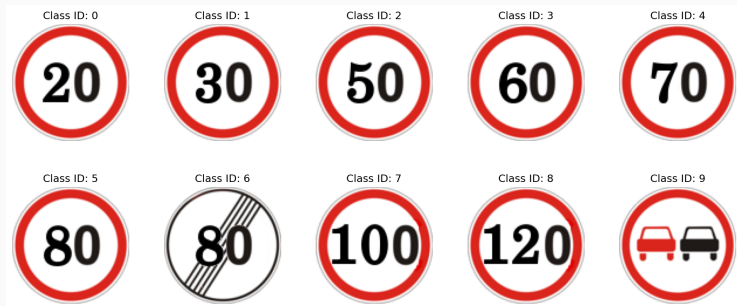
Dataset: GTSRB (10-class Subset)



- Over 10,000 images across 10 distinct traffic sign categories.
- Significant scale and illumination variations.

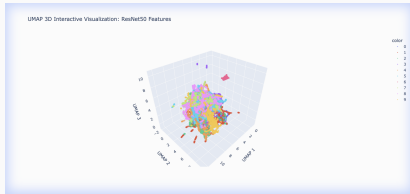
Methodology

Pipeline Architecture

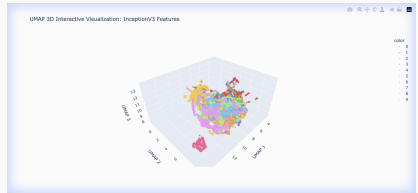


1. **Extraction:** ResNet50 & InceptionV3 backbones.
2. **Dimensionality Reduction:** PCA & UMAP (3D).
3. **Diagnostic Strategy:** Classification on 3D manifolds.

3D Manifold Visualizations (UMAP)



ResNet50 Manifold



InceptionV3 Manifold

UMAP preserves local structures, forming dense semantic clusters.

Results & Analysis

Qualitative Manifold Assessment

Metric	PCA (Linear)	UMAP (Non-linear)
Silhouette Score ($\uparrow$)	-0.139	-0.060
Davies-Bouldin Index ($\downarrow$)	11.126	6.022

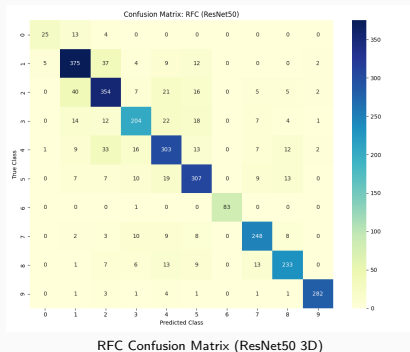
3D Manifold Separability Comparison

- **UMAP Advantage:** 45% improvement in DBI.
- **Non-linearity:** UMAP captures complex structures that PCA fails to isolate.

Classification in 3D Space

- **Insight:** Linear models fail in 3D (<35%).
- **Success: Random Forest** dominates by recursive partitioning.

3D Classifier	Acc. (R50)
SGDC (Linear)	28.94%
SVM-RBF	51.74%
Random Forest	82.28%

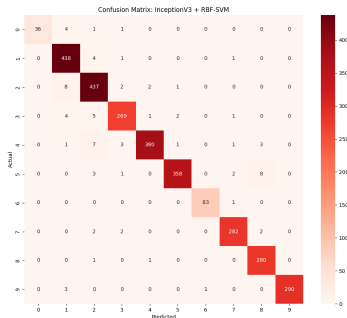


High-D Classification Results

Model	R50	IncV3
SGD (Lin)	91.2%	88.5%
RBF (Hyb)	95.9%	97.2%

Validation Acc.

RBF kernels effectively handle the density of deep manifolds, significantly outperforming linear baselines.



Peak Model CM (InceptionV3 + RBF)

Conclusion

- Hybrid architectures improve flexibility and performance.
- **RBF kernels** are essential for 2048-D features.
- 3D projections retain 82% accuracy, proving UMAP's density preservation.

GitHub Repository:

<https://github.com/toando0212/lab-m12>

Questions?